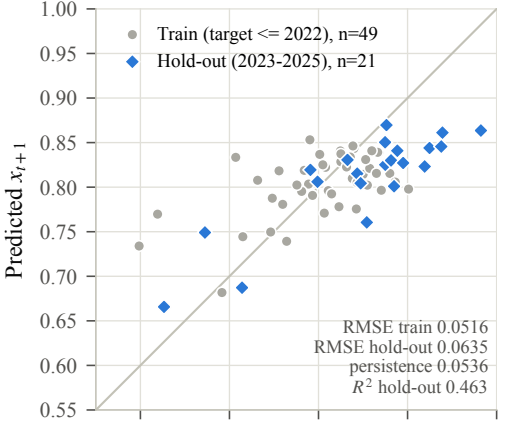
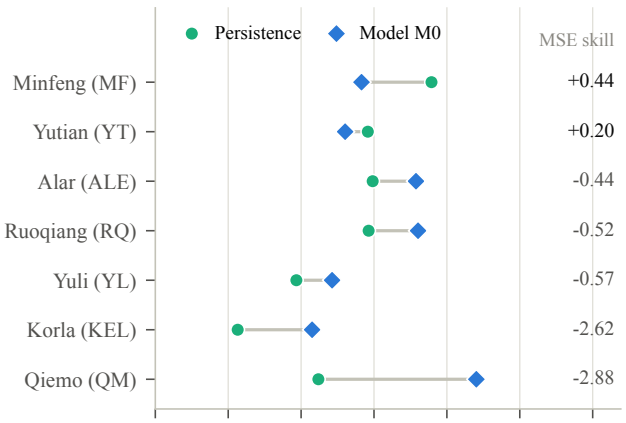

(a) One-step transfer, pooled r, d
Train (target <= 2022), n=49
Hold-out (2023-2025), n=21
RMSE train 0.0516
RMSE hold-out 0.0635
persistence 0.0536
R^2 hold-out 0.463
Predicted x_{t+1}
Observed $x_{t+1} = V/K$
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.6
0.7
0.8
0.9
1.0


(b) Hold-out error by site vs persistence
Persistence
Model M0
MSE skill
Minfeng (MF)
Yutian (YT)
Alar (ALE)
Ruoqiang (RQ)
Yuli (YL)
Korla (KEL)
Qiemo (QM)
+0.44
+0.20
-0.44
-0.52
-0.57
-2.62
-2.88
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Hold-out RMSE (V/K), n = 3 per site